

Color Routing and Beam Steering of Single-Molecule Emission with a Spherical Silicon Nanoantenna

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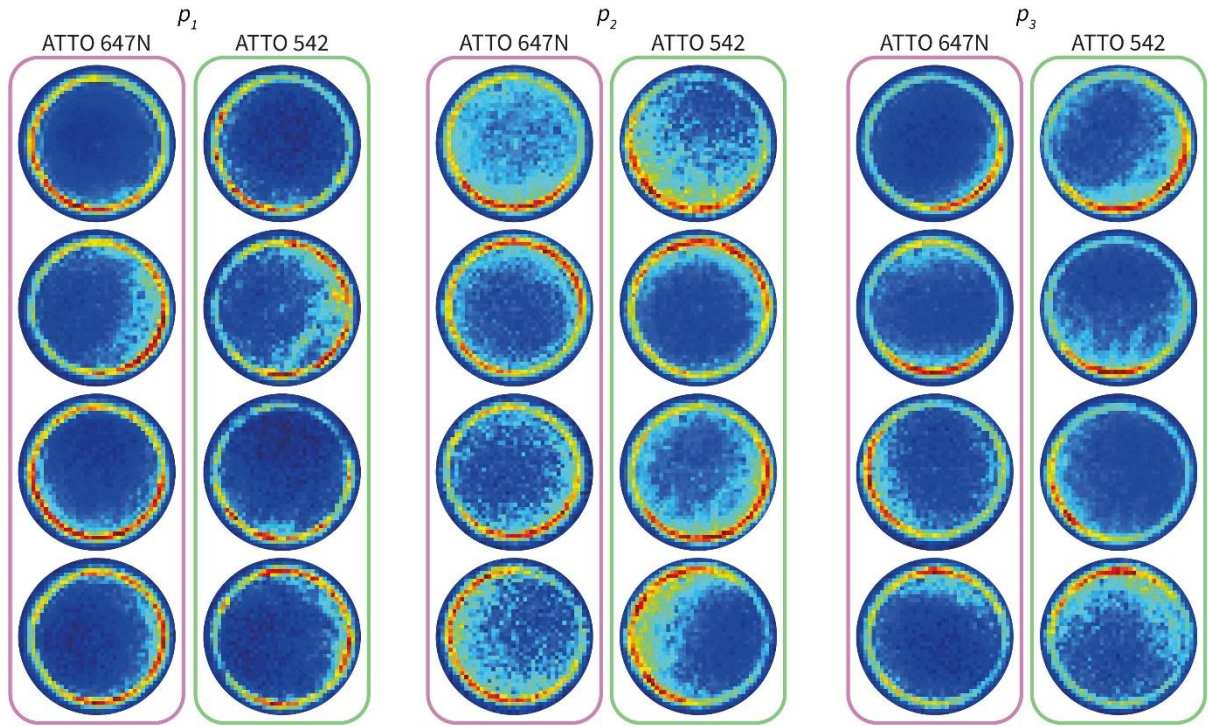


Figure S1: Broadband directional emission with a single 120 nm SiNP. Exemplary BFP images of ATTO 647N and ATTO 542 emission corresponding to the same structure, for the fluorophore pair positioned at distances p_1 (left), p_2 (middle) and p_3 (right) away from a 120 nm SiNP. Both fluorophores emit in the same direction.

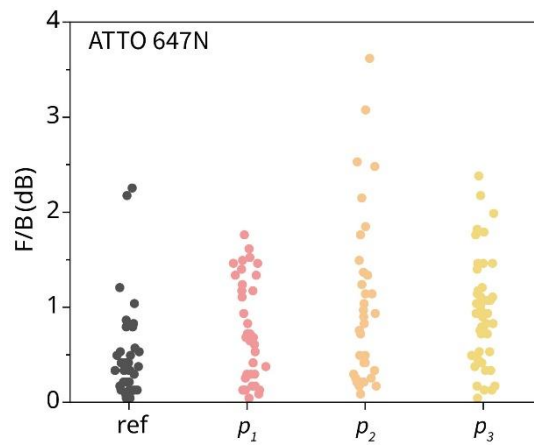


Figure S2: F/B quantification for 120 nm SiNP. Bee swarm distribution plots of the forward-to-backward (F/B) ratios obtained from the BFP patterns of ATTO 647N alone (ref) and at different distances p_i to a 120 nm SiNP.

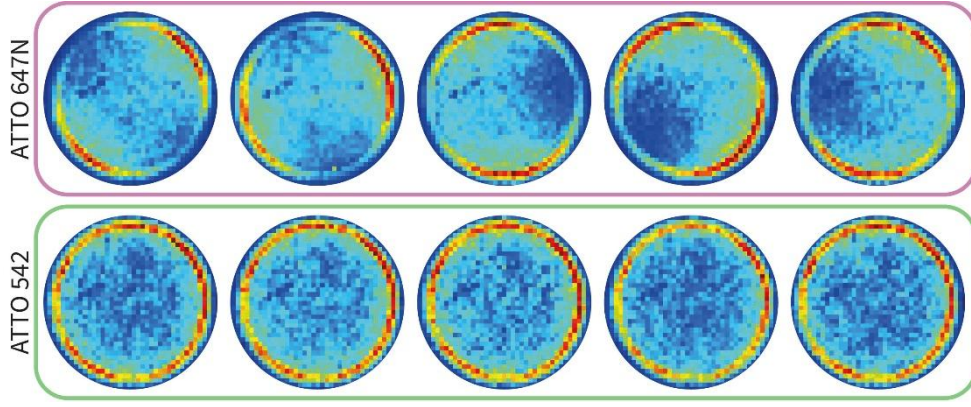


Figure S3: BFP images in the absence of SiNPs. Exemplary BFP images of ATTO 647N (top) and ATTO 542 (bottom) emission for origami structures in the absence of SiNPs (reference). ATTO 647N shows always a dipolar pattern, whereas ATTO 542 displays an isotropic emission characteristic of a rotating dipole.

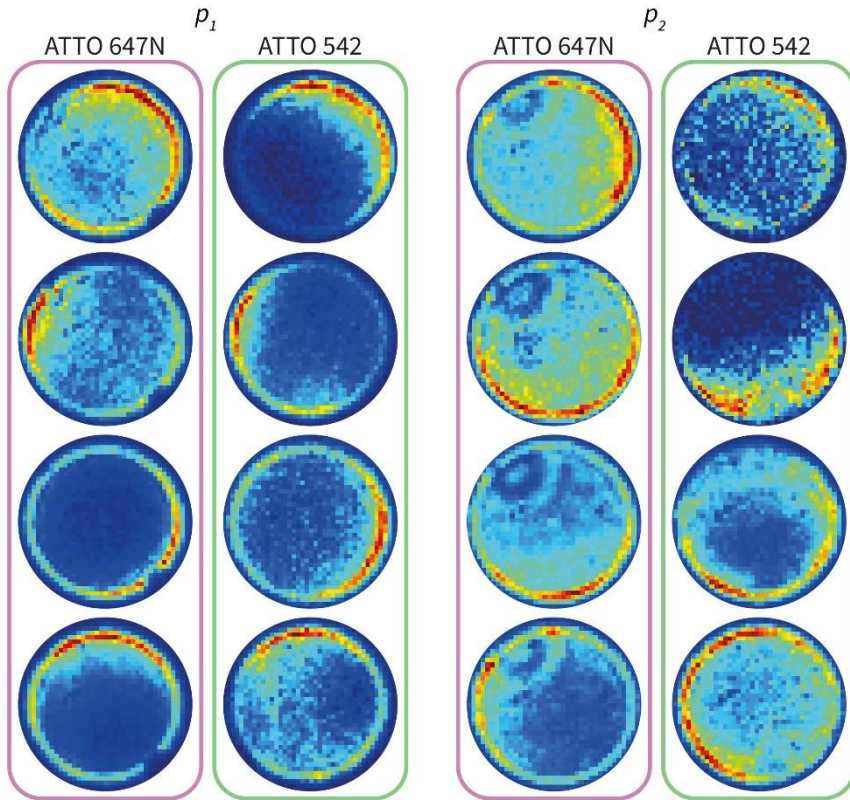


Figure S4: Broadband directional emission with a single 140 nm SiNP. Exemplary BFP images of ATTO 647N and ATTO 542 emission corresponding to the same structure, for the fluorophore pair positioned at distances p_1 (left) and p_2 (right) away from a 140 nm SiNP. Both fluorophores emit in the same direction.

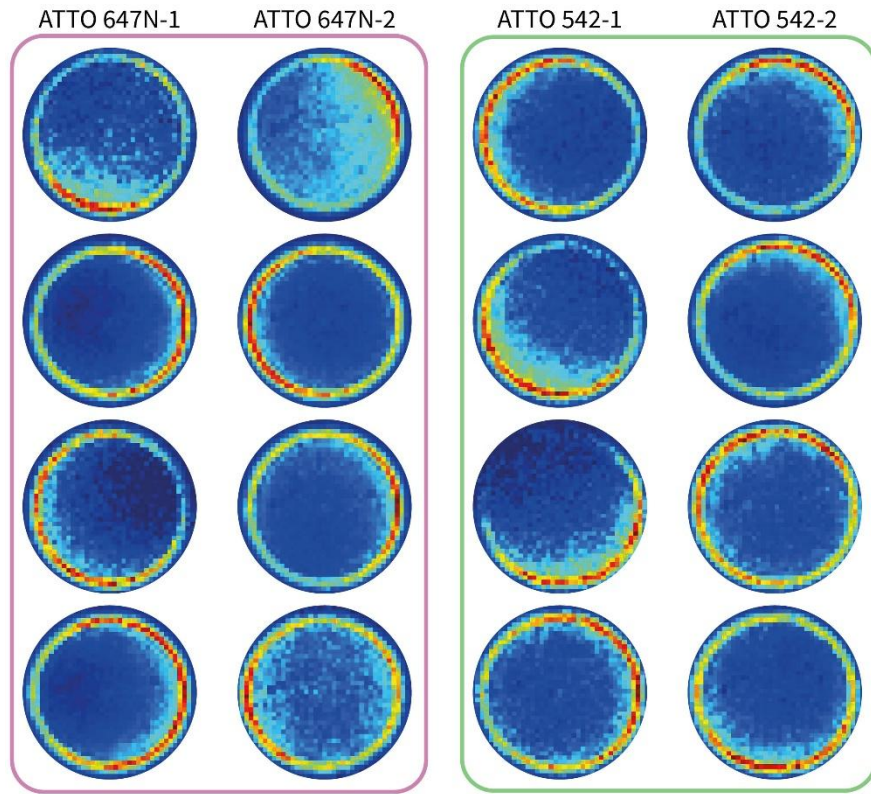


Figure S5: Beam steering with a single SiNP. Exemplary BFP images of two ATTO 647N (left) or two ATTO 542 (right) placed at opposite sides $p_{\pm 2}$ of a 120 nm SiNP. For each structure, the two fluorophores were labeled 1 and 2 according to their bleaching time. They emit into opposite directions.

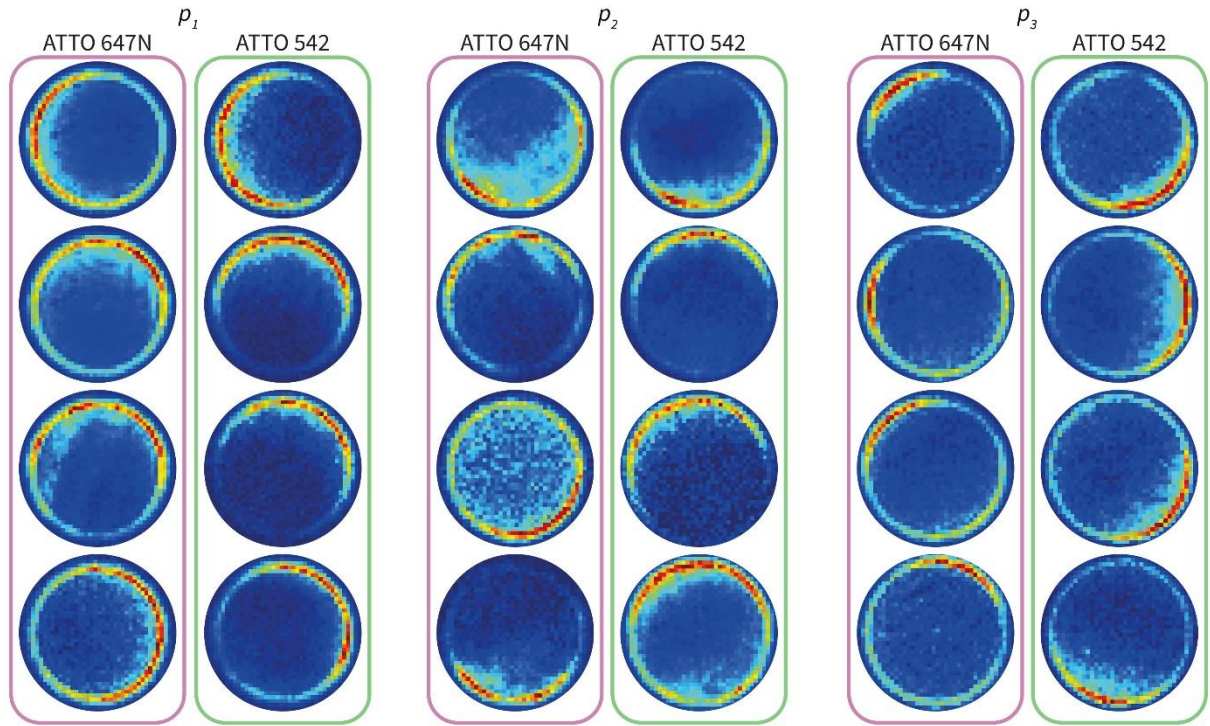


Figure S6: Color Routing with a single 170 nm SiNP. Exemplary BFP images of ATTO 647N and ATTO 542 emission corresponding to the same structure, for the fluorophore pair positioned at distances p_1 (left), p_2 (middle) and p_3 (right) away from a 170 nm SiNP. Color routing is observed for p_3 .

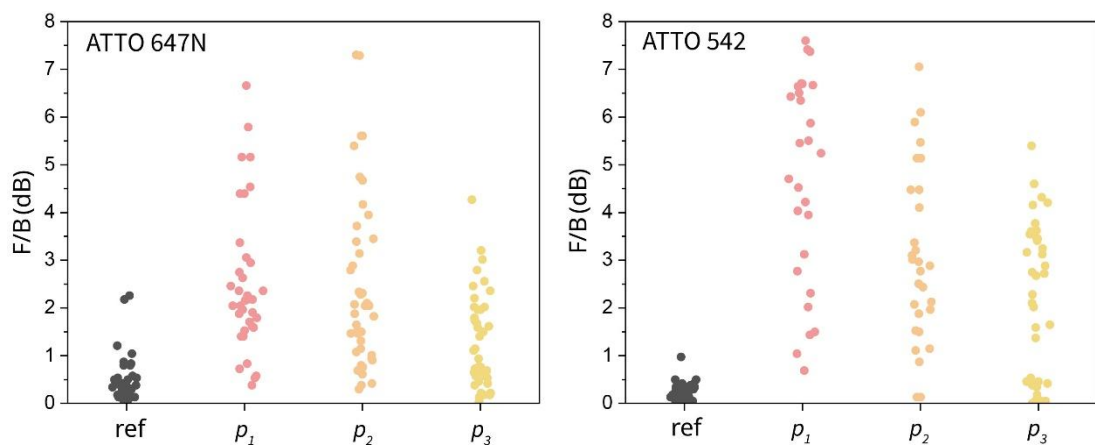


Figure S7: F/B quantification for 170 nm SiNP. Bee swarm distribution plots of the forward-to-backward (F/B) ratios obtained from the BFP patterns of ATTO 647N (left) and ATTO 542 (right) alone (ref) and at different distances p_i to a 170 nm SiNP. Values are higher compared to those in Figure 2D and Figure S3.

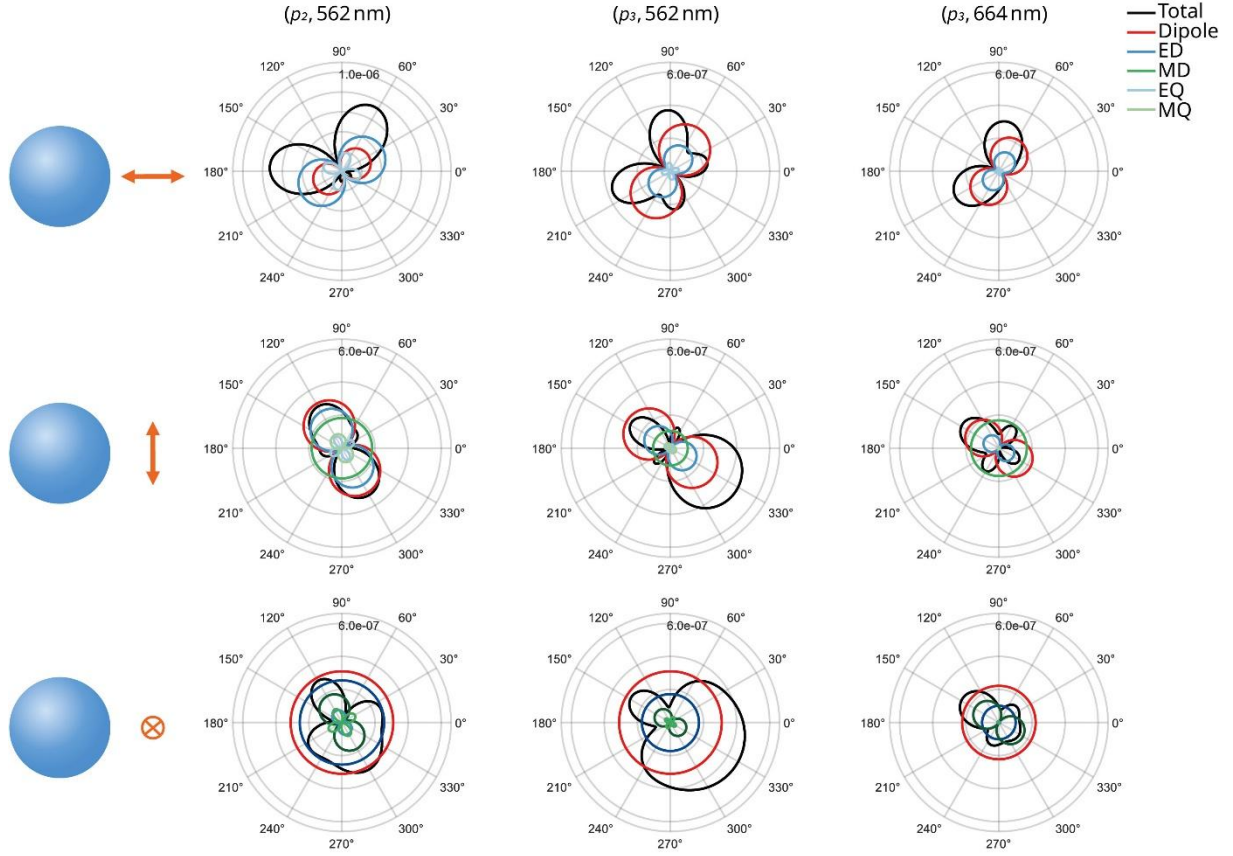


Figure S8: Multipolar decomposition. Multipolar decomposition of the far-field radiation patterns of a 170 nm SiNP excited by a dipolar source. Patterns are shown at different distances (p_2 and p_3) and emission wavelengths (562 and 664 nm) for the three possible orientations of the dipolar source (yellow arrow). The red line represents the dipole emitter, while ED, MD, EQ and MQ account for the induced multipoles in the SiNP.

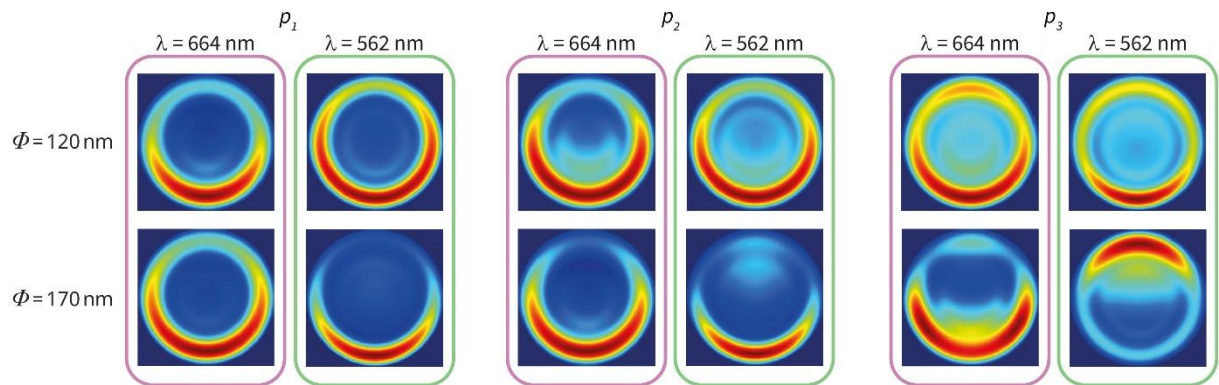


Figure S9: BFP simulations. Simulated BFP patterns for a SiNP of diameter 120 nm (top) and 170 nm (bottom) excited by a dipolar emitter (average orientation) located at positions p_i from the SiNP surface. Patterns are shown for the emission wavelengths representing the two dyes used in our experiments. These patterns are used to compute F/B plots shown in Figure 4.

Description of phasor diagrams in Figure 6D

For the case (p_2 , 562 nm), as the longitudinal orientation suppresses the pure forward and backward directions, the angles 180° and 270° have been selected as forward and backward, respectively. As explained in the main text, the longitudinal orientation is characterized by electric resonances only, where a constructive interference between ED and EQ, together with the emitter radiation, gives a strong forward radiation (collector regime), and a destructive interference of the emitter with the ED, and the ED with the EQ, heavily suppresses the backward direction.

On the other hand, for the p_3 cases, forward and backward directions can be defined more easily as 150° and 300° , respectively. For a wavelength of 562 nm, we observe that the backward direction features a similarly constructive interference between the emitter radiation and the ED, with a partially constructive contribution from the MD. The constructive interference between emitted and ED does not occur at the forward direction, with a slightly-destructive contribution from the MD, leading to the overall backward, reflector regime.

Finally, the 664 nm wavelength at p_3 showcases a similar phasor diagram in the forward direction. In this case, however, the ED is mostly constructive with the emitter radiation, with the MD not having a particularly strong interference with the other modes. This situation is reversed in the backward direction, where it can be seen that the MD almost perfectly destroys the emitter radiation. This leads to a sort of generalized Kerker condition, where the backward radiation is heavily suppressed. Thus, the color-routing effect can be understood as a result of constructive interference between emitter and induced ED modes in one direction, with the MD displaying a destructive interference with the emitter in the other direction. The direction at which this happens depends on the wavelength, yielding the color-routing behavior.

Name	Sequence
Biotin1	ACAGGAAGATTGTCCCCCTTATTCACCCCTCATTTGTTTC-Biotin
Biotin2	GTTGATAGATATAAGCATAAGTATAGC-Biotin
Biotin3	AGAGTACTCACGCTAACCTTTAATTGC-Biotin
Biotin4	CACTAAAACACTCACGAACTAACACTAAAGT-Biotin
Biotin5	TCACGACGTTGGGCGCTTTGGTAAAAC-Biotin
Biotin6	CAGAGATAGCGATAGTGAATAACATAA-Biotin
ATTO542_P1	AATTGAGTAATATCAGAAAATAAACAGCCATA-ATTO542
ATTO647N_P1	TACCCCGGTTAAAATTCCTTTGCCCGAACGTT-ATTO542
ATTO542_P2	ATTO542-CCCCCGCTGAGAGCCAGCAGGCCTGCAACAGTGCCA-CCCC
ATTO647N_P2	ATTATCACCGGAAATGTTAGCAAACGTAGAA-ATTO647N
ATTO542_P3	ATTATTACTTGGGAAGTTCATTACCCAAATCA-ATTO647N
ATTO647N_P3	ATTO647N-CCCCCAGAGCGGGAGCTAACTTTCCTCGTTAGAAT-CCCC
ATTO542_P-2	ATGCTGATGTTGGGTTTAGCTTAGATTAAGAC-ATTO542
ATTO647N_P-2	CCCCTACCGACAAAAGGTAATAAGAGAATATAAAGCCCCCTT-ATTO647N

Table S1: List of biotinylated and dye-modified staple sequences used in this project compared to the original design.^[39] ATTO 647N and ATTO 542 staples present a single ATTO 647N or ATTO 542 fluorescent molecule for the alternative positions (p_1 , p_2 , p_{-2} and p_3), and biotin staples allow binding to a neutravidin-functionalized glass surface.

Name	Sequence
H.left 1(A8)	TACAAATTGCCAGTAAAGTAATTCTGTCCAGAAAAAAAA
H.left 2(A8)	TAAAGACTGTTACTTAGGCGCAGACGGTCAATAAAAAAAA
H.left 3(A8)	GCCACCCTTCGATAGCATAATCCTGATTGTTTAAAAAAAA
H.left 4(A8)	CACCAACCAAGTACAAGTACAGACCAGGCGCAAAAAAAAA
H.left 5(A8)	CGACATTCCCAGCAAAATTATTTGCACGTAAAAAAAAAAA
H.left 6(A8)	CTTTTTTTTCATTTCAACAATAACGGATTTCGAAAAAAAA
H.left 7(A8)	ACGGGTAATAAATTGTTGACCAACTTTGAAAGAAAAAAAA
H.left 8(A8)	ATTAATTATGAAACAATATACAGTAACAGTACAAAAAAAA
H.left 9(A8)	AATATTGACGTCACCGTGCGTAGATTTTCAGGAAAAAAAA
H.left 10(A8)	AACAGTAGCCAACATGACATGTTTCAGCTAATGAAAAAAAA
H.left 11(A8)	AAGAACGCAAGCAAGCATAATATCCCATCCTAAAAAAAAA
H.left 12(A8)	CTTGCGGGGTATTAAAAAACCAATCAATAATCAAAAAAAAA
H.left 13(A8)	AAATCAGATCATTACCATCAACAATAGATAAGAAAAAAAA
H.left 14(A8)	CATATGGTAGCAAGGTAATGGAAGGGTTAGAAAAAAAAA
H.left 15(A8)	CCATCTTTTCGTTTTCAAACCACCAGAAGGAGCAAAAAAAAA
H.left 16(A8)	GAGCCACCATCAAGTTTCTGATTATCAGATGAAAAAAAA
H.left 17 (A18)	AAAAAAAAAAAAAAAAGGATTATACTTCTGAACCGGAAACGTCACCAA
H.left 18 (A18)	AAAAAAAAAAAAAAAAACCTACCATATCAAAATCACCAGTAGCACCA
H.left 19 (A18)	AAAAAAAAAAAAAAAAATTTACGAGCATGTAGCCAAGTACCGCACTCA
H.left 20 (A18)	AAAAAAAAAAAAAAAACATAAGGGAACCGAACGTCGAAATCCGCGACC
New Core 1	GAAACGCAAAGACACCGCCAAAGATACCGAAG
New Core 2	ATTTTGCACCCAGCTATTAGCGAAAGAATTAA
New Core 3	AGCGAAAGACAGCATCAGGAAGTTTGTAGCAT
H.left 22(A18)	AAAAAAAAAAAAAAAAGCCGCCACTCATCAATAGCACCGTAATCAGT
H.left 23(A18)	AAAAAAAAAAAAAAAAATGGCAATCAGAACCCGCCTCCCTCAGGAGG
H.left 28(A8)	TGAGCGCTTAAGCCCATGGCATGATTAAGACTAAAAAAAA
H.left 29(A8)	CTGAACACATAGCAATAAACGCAATAATAACGAAAAAAAA
H.left 30(A8)	AAACAGGGAAGAAAAGGCCGAACAAAGTTACAAAAAAAA
H.left 31(A8)	TCCACAGAGGTGTATCGGATAAGTGCCGTCGAAAAAAAA
H.left 32(A8)	GTCACCAGCGCCACCCTTAGGATTAGCGGGGTAAAAAAAA
H.left 33(A8)	TAGGAACCCACCCCTCTATTAAGAGGCTGAGAAAAAAAAA
H.left 34(A8)	AATTAGAGAACCGATTGGCAACATATAAAAAAAAAAAA
H.left 35(A8)	TTACCATTTTACCAGCACGGAATAAGTTTATTTAAAAAAAA
H.left 36(A8)	TGAAACCACAGAACCGAGCCACCACCCTCAGAAAAAAAAA
H.left 37(A8)	GCGACAGAACCGGAACACCACCAGAGCCGCCGAAAAAAAA

Table S2: List of DNA modified staples, which were extended as NP capturing strands, used for the directional antennas compared to the original origami design.^[39] Handles are staples extended with an PolyA (number of A indicated in parenthesis) to bind a single PolyT functionalized SiNP.

		$\Phi = 120$ nm	$\Phi = 170$ nm
i	x_i (nm)	$ d_i $ (nm)	$ d_i $ (nm)
-2	-50	20	15
1	-25	8	6
2	50	20	15
3	133	85	75

Table S3: List of fluorophore positions in the DNA origami (taking the SiNP as $x = 0$) and estimated distance d_i to the SiNP surface depending on its diameter.